

RDS or Radiative Dust Sensor is an attachment to MEDA on the Mars 2020 rover, designed to measure solar radiation at different wavelengths due to the scattering of dust particles. RDS is found to be a particularly useful tool onboard MEDA, capable of detecting dust vortices or dust lifting events (wind gusts or dust storms). Because there are several sensors placed at different locations on the RDS, the Top and Lat channels can infer more variables from the data. This includes the diameter of a dust devil (by using the FOV of the Top and Lat sensors) and also the DD's trajectory. Now, you may be wondering what units are used in a graphical analysis of RDS data. All of the RDS graphs in the Toledo paper used RDS Normalized Signals vs. Time (min. or sec.). I'm not sure what an RDS Normalized Signal is, but my research indicates the recommended units are irradiance vs. time. Irradiance is the amount of solar radiation reflected off the sensors.

I do not believe the sampling rate is a concern for the perseverance data, as the rover has an established 1 Hz sampling frequency or 1 samples/s; any less increases the chance of error in detection. Regarding any potential missing values, I do not think there are any at this moment. The RDS-calibrated data in the PDS includes data from all affected sensors and the time of impact. There are 1726 sols of available RDS data within the PDS, which is where the authors of the Toledo paper acquired their martian RDS data.